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# Conversation:
# Playable systems: Game co-design for systemic intervention

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## 1. The organising research questions

> In the face of systemic problems, how might game co-design scaffold the development and sharing of knowledge to unravel complexity, generate insights, and facilitate change in organisational settings?

This organising research question is based on the recent critiques of systemic design approaches, from the perspectives of pragmatism (Vink, 2023) and visual communication (Hvidsten & Kirah, 2023; Julier & Kimbell, 2019).

Further, we continue with the following sub-questions, related to the organising research question and the “Playable Systems” approach:

a) How might rapid prototyping of games be effectively deployed in conversations around organisational and social complexity?

b) What approaches (elements, components, mechanics, etc.) are best for scaffolding the rapid prototyping of games with non-designers?

c) How do we capture social insights and structural innovations uncovered during the various stages of a (game) design process? How do we move these into implementation?

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## 2. Context: Game design as systemic intervention?

This DRS conversation aims to open the discourse around game co-design as an experimental method for engaging with systemic problems and expanding current practices (and critiques) in systemic design by centering the discussion around the current version of the "Playable Systems Toolkit."

With the introduction of "Wicked Problems" (Rittel & Webber, 1973; Buchanan, 1992) as a design concern, complexity, and ways to address it in various design contexts, have been at the forefront of design research and practice. Some tools have been developed, and/or appropriated from other disciplines, to help manage complexity. In the so-called "systemic design" or "systems-oriented design" community, systems maps (e.g. Meadows, 2008), commonly used in systems engineering and other systems thinking disciplines (biology, sociology, etc.) have been widely deployed as productive tools for grappling with the complexity of design scenarios - particularly in disciplines such as service design. More recently these mapping approaches have been critiqued for their static representation of dynamic systems; employing an expert (rather than participatory) practice (Hvidsten & Kirah, 2023); lack of transparency, and their unproven value in suggesting and testing appropriate interventions. These limitations are consistent with acknowledged scaling problems in design-driven innovation projects; namely, they fall short when moving from a single project to a wider systems level (Almqvist, 2017; 2020).

In efforts to accommodate these shortcomings, the Playable Systems Group has been developing a set of game design practices centred around flexible co-design structures (game elements and workshop instructions). Similar to work that attempts to use game design to introduce and facilitate participatory design workshops (Brandt & Messeter,2004; Vaajakallio & Mattelmäki, 2014), our approach suggests that game design can be democratised and deployed to create systemic insights among diverse stakeholders about their specific situation, rather than simply to aid in the co-design process. Our group suggests that the main aspect of co-design is learning together, over time, through multiple (and sometimes contradicting) perspectives. Games provide effective ways to facilitate learning, encourage critical thinking and increase understanding and insight in ways that are difficult to achieve with traditional methods and tools (Caruson, 2005; Shellman & Turan, 2006; Alme & Hvidsten, 2022). The Playable Systems Kit, as we have called it, contains blank and empty game objects, such as cards and dice, that, when combined with a scaffolded game design process, can be used to create playable game prototypes that allow for more dynamic exploration of design scenarios, enable a safe space for discussion and critique for stakeholders, and the design and prototyping of solutions to those issues.

## 3. Session set-up: A hybrid conversation

In this hybrid session, a link gives participants access to materials for discussion, including imagery, description, and a printable example of the current version of our Playable Systems

Kit. This is so they can explore the design elements in the kit on their own, and/or as part of the conversation.

Due to a variety of factors (small room size, larger number of participants, technical issues with room microphone) we altered the initial schedule shown below and removed the hands-on component to the conversation. Participants were still encouraged to use the materials on their own time and to contact the hosts with questions. We began by discussing three exemplars from our work in playable systems as described in Section 5.

> 0:00-0:10: The session begins with distributing the Playable Systems Kit (blank board, custom dice, empty cards, etc.). We will bring as many copies as possible, so participants can explore and play while waiting for all participants to arrive.
>
> 0:10-0:30: The session will then proceed with a discussion of the above context as well as the current version of instructions for using the Kit in participatory design activities.
>
> 0:30-1:10: Participants will form groups and discuss one or more of the Kit elements in the context of a design problem they collectively decide. Rather than asking participants to use the Kit elements extensively, they will prompt discussion about the value of gamification for co-design in various settings, how to best elicit good results, and how they could see the Kit or other aspects of game design (mechanics, characterization, etc.) being leveraged effectively to support systemic change.
>
> 1:10-1:30: The session will end with comments, questions, and open discussion about the Kit itself, as we seek feedback and collaborators for future iterations.

## 4. Conversation summary

We opened the session by introducing the concept of systems mapping to help participants understand where Playable Systems fits into the current discourse. We followed by describing examples of Playable Systems. The first was the Systems of Vacancy workshop conducted by Dr. Barba's group in Washington DC which outlined the workshop design process and implementation to showcase the methodology. The second was a series of workshops integrated into coursework at Kristiana University College in Oslo led by Dr. Hvidsten. In one series of workshops, students co-designed board games around complex problems, given by clients, as part of the students' design research process. While generating discussions across stakeholders, game design competency was a challenge as students had no prior experience. In another series of workshops, students were able to map a complex everyday problem using the familiar game of Monopoly. Here, they were able to complete the task without prior game design experience. A third example, presented by professor Kirah, demonstrated the value of embodied play for representing and communicating systems dynamics through a field study of game design used to teach financial literacy to participants in Uganda. Dr. Vink provided an example of student-designed games used to represent interactions on mental health wards in Oslo hospitals to make the point that while the games were able to effectively represent the system dynamics, a point validated by stakeholders, they also robbed those same stakeholders of agency because they were not involved in a co-design process. Dr. Rouse concluded the introduction by critiquing an underlying assumption in the examples by pointing out that

abstracting stakeholders and system dynamics into game components and rules has limitations (i.e., stakeholder design expertise, time, power dynamics) which can be mitigated with more embodied forms of play, such as role playing — also reminding us the "The best game is game design."

The open discussion saw participation from many of those present, and interaction and discussion continued after the allotted time. A few major themes surfaced:

## *4.1 Participation and inclusivity*

Although not discussed during the introduction, many in the room had questions and comments around how dependent Playable Systems' success might be on who is doing the designing. As one participant put it:

> "When or how do you go about making sure you've got the right people in the room to be able to get to a point where you've kind of played through the scenarios and might have a way forward?"

Participant selection was then discussed in the context of the examples given. While the co-design process described was determined to be helpful at mitigating some of these concerns it was ultimately pointed out that there is no silver bullet for participant selection. Designers should take great pains to ensure that representation is intentional and diverse, but ultimately the outcome is subject to forces outside the designers' control (client needs, time, money, location, etc.). It was pointed out that Playable Systems were no better or worse than other co-design methodologies in this regard, and furthermore all science research is subject to this kind of sample bias. So in many ways the question was orthogonal to what was under discussion.

## *4.2 Safe and brave spaces*

The potential for Playable Systems to open up dialogue on difficult questions and potentially invite or mitigate confrontation was discussed at some length. This part of the discussion was initiated by a participant:

> "It can also be interesting to work on taboo subjects that are hard to talk about. Which is my subject of research right now. I'm working with women that have injuries after birth. And in the second study we are thinking to do co-design. That's the plan, but I am thinking now having seen [this] that making a game might make it more fun and make them more open to talk about it."

Since the beginning of our exploration into Playable Systems we have always held this possibility to be one of the strongest arguments for implementing the approach. It was pointed out that the conveners also have a strong inclination to believe this is the case but there is very little evidence to support this at the moment. This will be a major research question in the next phase of the work. Dr. Rouse mentioned that this is one area where roleplaying, particularly Nordic Larp, has a longstanding tradition of engaging with serious topics through play that should be considered when taking a Playable Systems approach.

## *4.3 New domains and getting started*

Many participants, particularly Ph.D. students, were curious about how to apply Playable Systems to their own work, and some pointers and caveats were offered by the conveners for different domains including: education, reproductive health, financial planning, conflict resolution, and the Finsbury Park project in London UK.

Many were also interested in how to start doing Playable Systems work without a background in Game Design, and the related topic of 'preparation' was also discussed with one participant asking:

> "When you have a game in place, how much information do we send in advance so that you don't, like, kill the fun factor, but that you inform the participants enough?"

In addition to offering participants access to the workshop materials discussed (email: evan.barba@georgetown.edu for information), conveners and participants suggested a number of texts and strategies for getting started or finding inspiration, including:

- Katie Salen and Eric Zimmerman. 2003. Rules of Play: Game Design Fundamentals. The MIT Press.
- The World Peace Game Foundation. https://worldpeacegame.org/the-game/
- Eric Zimmerman. 2023. The Rules We Break. Lessons in Play, Thinking, and Design. Princeton Architectural Press.

## *4.4 Reflections and limitations*

This conversation was well-attended and participants were extremely engaged. There was particular enthusiasm from Ph.D. students in the room working on problems in the domains cited above. One critique of the Playable Systems approach that emerged from the discussion was the question of feasibility. In contrast to other co-design methods which can often be transferred and applied quickly and without requiring any advance or specific training of participants, game design is quite a bit more challenging. The requirement to take stakeholders' embodied and experienced knowledge of a system and create abstract rules, representations, and interactions in a game world is not a natural skill for many people. The role of the designer as facilitator in this case is extensive and has the potential to bias the outcome perhaps more than other co-design methods.

The biggest challenge identified by the group was time. Producing rich and nuanced games takes many iterations and without real dedication by stakeholders, game design can be seen as too intensive and burdensome to be applied widely. This is as much a call for more lightweight, effective, and portable game design methods as it is a caution for practitioners within a Playable Systems approach. Ideally, game co-design is just one step in a much longer process of development and refinement. Getting buy-in from clients, stakeholders, and participants to continue to develop, play, and iterate on games is likely to be difficult but also a rewarding way of unlocking the full potential of games as Playable Systems.

## 5. References

Alme, V., & Hvidsten, A. (2022). To Learn or not to Learn: On the Importance of Mode Switching in Educational Wargames. *Scandinavian Journal of Military Studies, 5*(1), 269–281.DOI: https://doi.org/10.31374/sjms.123

Almqvist, F. (2017). The fuzzy front-end and the forgotten back-end: User involvement in later development phases. *The Design Journal, 20*(1),.2524-2533.

Almqvist, F. (2020). *Service Design in the Later Phases* [Doctoral Thesis, The Oslo School of Architecture and Design]. ADORA. https://hdl.handle.net/11250/2657504

Brandt , E. and Messeter , J. (2004). Facilitating collaboration through design games. In *Proceedings of the 8th Conference on Participatory Design: Artful Integration: Interweaving Media, Materials and Practices*, vol. 1, pp. 121–131, New York: ACM Press. DOIi: 10.1145/1011870.1011885.

Caruson, K. (2005). So, You Want to Run for Elected Office? How to Engage Students in the Campaign Process Without Leaving the Classroom. *PS: Political Science & Politics, 38*(April), 305–310.

Buchanan, R. (1992). Wicked problems in design thinking. *Design issues*, *8*(2), 5-21.

Hvidsten, A. & Kirah, A. (2023). Relics and Resources: Representing complexity in service and systemic design. In *Proceedings from the RSD5 Symposium*. ISSN 2371-8404.

Julier, G., & Kimbell, L. (2019). Keeping the System Going: Social Design and the Reproduction of Inequalities in Neoliberal Times. *Design Issues, 35*(4), 12-22. DOI:10.1162/desi_a_00560

Kirsikka Vaajakallio & Tuuli Mattelmäki (2014) Design games in codesign: as a tool, a mindset and a structure, *CoDesign, 10* (1), 63-77, DOI: 10.1080/15710882.2014.881886

Meadows, D. H. (2008). *Thinking in systems: A primer*. Chelsea Green Publishing.

Rittel, H. W., & Webber, M. M. (1973). Dilemmas in a General Theory of Planning. P*olicy sciences, 4*(2), 155-169.

Shellman, S. M., & Turan, K. (2006). Do Simulations Enhance Student Learning? An Empirical Evaluation of an IR Simulation. *Journal of Political Science Education, 2*(1), 19–32. https://doi.org/10.1080/15512160500484168

Vink, J. (2023). Embodied, Everyday Systemic Design- A Pragmatist Perspective. *Design Issues, 39*(4), 35-48. https://doi.org/10.1162/desi_a_00731

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**Anna Kirah** As a design anthropologist and psychologist, Anna is an expert in qualitative research methods used to understand the needs of people and to connect this understanding to the creation of meaningful, relevant, useful and desirable products, services and organisational change.

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